

VRDots Cueing Effect Ranking — Clean Experiments Only

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Ranked by Δpp (CUED – UNCUEd percentage points). All experiments with any artifact flag are excluded. Chance = 12.5% (1/8 directions). Single-session depth results have large standard errors ($\sim \pm 15$ pp at $n = 64/\text{arm}$).

Depth Experiments — Clean Conditions Only

The depth artifact (transform.forward pitch bias) only applies when depth planes **swap at tStart**. DepthBaseline and DepthParam have no swap at tStart — they are fully clean regardless of session date.

Rank	Experiment	Condition	Δpp	Notes
1	DepthBaseline 0.10 m	Far plane	$\sim +62$ pp	Avg across S1 (+59 pp) and S2 (+65 pp); $n \approx 32/\text{cell}$ per session
2	DepthParam all depths	Far plane	+47 pp	Consistent across 0.03–0.15 m disparity; disparity-independent
3	DepthParam 0.03 m	Near plane	+13 pp	Only positive Near result
—	DepthParam 0.05 m	Near plane	–9 pp	Cueing inverts
—	DepthParam 0.10 m	Near plane	–22 pp	Cueing inverts
—	DepthParam 0.15 m	Near plane	–25 pp	Cueing inverts
—	DepthBaseline 0.10 m	Near plane	–26 pp	Strongly inverted; high session variance (–5 vs –47 pp across 2 sessions)

Key pattern: Far-plane cueing ($\sim +62$ pp) is the largest clean effect in the entire dataset — nearly double the 2D baseline. It is disparity-invariant: Far cueing locks at $\sim +47$ pp regardless of depth separation. Near-plane cueing inverts at all but the smallest disparity tested.

The Far-plane boost over 2D is approximately **+27 pp** ($62 - 35 \approx 27$ pp), suggesting that binocular disparity adds a large independent contribution on top of the temporal onset cue.

2D Experiments — All Clean

Rank	Experiment	Ap radius	Dots/field	Condition	Δpp	CUED%	UNCUEd%
1=	Density 63 dots	3.5°	63	N	+34.8 pp	60.5%	25.8%

1=	Density 500 dots	3.5°	500	N	+34.8 pp	63.3%	28.5%
3	Density 173 dots	3.5°	173	N	+33.6 pp	58.6%	25.0%
4	DepthSwapCtrl N (binocular)	3.5°	63	N (no swap)	+34 pp	—	—
5	Density 1000 dots	3.5°	1000	N	+25.0 pp	53.1%	28.1%
6=	DecoupledDots post-fix	3.5°	63	N (no swap)	+23.5 pp	48.4%	24.9%
6=	DecoupledDots post-fix	3.5°	63	C (color swap only)	+23.5 pp	50.0%	26.5%
8=	S&B Replication	2.0°	63	MC (motion+color swap)	+19.9 pp	70.7%	50.8%
8=	NMoCol (Çatak params)	1.65°	43	MC (motion+color swap)	+19.9 pp	—	—
10=	S&B Replication	2.0°	63	N	+16.8 pp	65.2%	48.4%
10=	NMoCol (Çatak params)	1.65°	43	N	+16.8 pp	—	—
12	NMoCol	1.65°	43	M (motion swap only)	TBD	—	—
13	NMoCol	1.65°	43	C (color swap only)	TBD	—	—
14	Çatak MCvsDb	1.65°	43	N	+14.6 pp	—	—
15	Çatak MCvsDb	1.65°	43	MC	+11.5 pp	—	—
16	Çatak MCvsDb	1.65°	43	Db	+10.4 pp	—	—

OR = odds ratio (baseline-corrected effect size). TBD = tested but script re-run needed for NMoCol M/C individual conditions.

Key Findings from the Ranking

1. Aperture / dot count is the dominant 2D variable. The large aperture (3.5° radius, 63–500 dots) gives ~~+35 pp — twice the Çatak aperture effect (+15 pp)~~ and double S&B's original group result (+20 pp). This is

not density: the effect is flat across 8× dot-count range (63–500 dots, $OR \approx 4.3\text{--}4.4\times$). It collapses only at 1000 dots ($OR = 2.9\times$).

2. Cueing survives all 2D swaps tested at large aperture. At Ap 3.5°, neither motion swap, color swap, nor their combination disrupts cueing — the DecoupledDots C condition (+23.5 pp) matches N (+23.5 pp). At Ap 1.65°, MC reduces cueing only modestly (~+12 pp vs +16 pp baseline). The cue is robust to signal identity changes.

3. Far-plane depth triples the cueing effect. Far-plane cueing (~+62 pp) far exceeds anything achievable in 2D. The Near plane inverts cueing at all but the smallest disparity, producing a striking asymmetry that is present in binocular but absent in monocular sessions (DepthSwapCtrl clean N condition), implicating stereoscopic processing.

4. Depth-field continuity boosts cueing beyond dot cueing alone. From DecoupledDots post-fix GLM: dot cueing alone (F1) = +22.3 pp ($OR = 3.07\times$); depth-field cueing alone (F2) = +12.5 pp ($OR = 1.89\times$); both together = much larger. The conjunction is required — neither factor alone reaches the Far-plane level.

Excluded (Artifact Flags)

Experiment	Sessions	Problem	Excluded conditions
DepthSwapCtrl ZdA/ZdB	All (pre-fix)	transform.forward pitch bias (attenuated, ~26% UP)	ZdA, ZdB
DepthColorLinked ZdNoi/ZdCoh	All (pre-fix)	Same, attenuated	ZdNoi, ZdCoh
DecoupledDots pre-fix Z/CZ	260406–260407	Full artifact (50% UP in wrong responses)	Z, CZ
DecoupledDots sign-bug session	260415_2242	Disparity sign inverted	Near/Far breakdown only

The ZdA collapse and ZdCoh disruption results — the primary object-based specificity finding — are excluded from this ranking pending replication with clean `_v2` assets.